

RMO– 2023 KV

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Given a triangle ABC with $\angle ACB = 120^\circ$, The point L is marked on the side AB such that CL is the bisector of $\angle ACB$. The points N and K are marked on the sides AC and BC, respectively so that $CN + CK = CL$. Prove that the triangle KLN is equilateral.
2. Given a prime number p such that the number $2p$ is equal to the sum of the squares of some four consecutive positive integers. Prove that $p - 7$ is divisible by 36.
3. Let $f(x)$ be a polynomial with real coefficients of degree 2. Suppose that for some pairwise distinct real numbers a, b, c we have: $f(a) = bc$, $f(b) = ca$, $f(c) = ab$. Determine $f(a+b+c)$ in terms of a, b, c.
4. The set X of N four-digit numbers formed from the digits 1, 2, 3, 4, 5, 6, 7, 8 satisfies the following condition:

For any two different digits from 1, 2, 3, 4, 5, 6, 7, 8 there exist a number in X which contains both of them.

Determine the smallest possible value of N.

5. The side lengths a, b, c of a triangle ABC are positive integers. Let:

$$T_n = (a + b + c)^{2n} - (a - b + c)^{2n} - (a + b - c)^{2n} + (a - b - c)^{2n}$$

for any positive integer n. If $\frac{T_2}{2T_1} = 2023$ and $a > b > c$, determine all possible perimeters of the triangle

ABC.

6. The diagonals AC and BD of a cyclic quadrilateral ABCD meet at P. The point Q is chosen on the segment BC so that PQ is perpendicular to AC. Prove that the line joining the centres of the circumcircles of triangles APD and BQD is parallel to AD.

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